

Progress Test 3.A — Elec-S1

3.A · Kirchhoff's laws (combinations, divider, Millman)

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Equivalent resistance

(1 mark)

Give the equivalent resistance of $R_1 = 6.0 \text{ k}\Omega$ and $R_2 = 3.0 \text{ k}\Omega$ connected in parallel.

$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2} = \frac{6.0 \times 3.0}{6.0 + 3.0} \text{ k}\Omega = \frac{18}{9.0} \text{ k}\Omega$	$R_{\text{eq}} = 2.0 \text{ k}\Omega$	(smaller than the smaller one)
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Marking

①
0 1

Marking scheme

① 1 pt $R_{\text{eq}} = 2.0 \text{ k}\Omega$ (“product over sum”).

Exercise 2. Kirchhoff's laws

(5 marks)

State Kirchhoff's node law and mesh law. For each law, give the fundamental principle it expresses.

Node law. At any node, the sum of the currents entering the node equals the sum of the currents leaving it: $\sum i_{\text{in}} = \sum i_{\text{out}}$, i.e. $\sum_k \varepsilon_k i_k = 0$ with $\varepsilon_k = +1$ for a current entering and -1 for a current leaving. Mesh law. Along any mesh (closed path), the algebraic sum of the voltages is zero: $\sum_k \varepsilon_k u_k = 0$, with $\varepsilon_k = +1$ if the arrow of u_k points in the direction of travel and -1 otherwise. The node law expresses the conservation of electric charge (no charge piles up at a node). The mesh law expresses the conservation of energy (after a closed path the potential returns to its starting value).
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Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts Node law: statement (1 pt) and algebraic form with signs (1 pt).
 ② 2 pts Mesh law: statement (1 pt) and sign convention along the path (1 pt).
 ③ 1 pt Charge conservation *and* energy conservation, each matched to its law.

Examiner's tip

Box 6 of the handout is easy to misread: its brackets refer to the *first* sentence of each paragraph. The same current through components in series comes from the node law; the same voltage across components in parallel comes from the mesh law. Currents add up at a node (node law); voltages add up round a loop (mesh law).

Exercise 3. Quick circuit laws

(3 marks)

$R_1 = 1.0 \text{ k}\Omega$ and $R_2 = 3.0 \text{ k}\Omega$ are in series under $u = 12 \text{ V}$. The voltage u_2 across R_2 is:

- ☐ 3.0 V ☐ 4.0 V ☒ 9.0 V ☐ 12 V

Why: voltage divider: $u_2 = \frac{R_2}{R_1 + R_2} u = \frac{3}{4} \times 12 = 9.0 \text{ V}$. The larger resistance takes the larger share.

A circuit has $N = 3$ electrically distinct nodes and $B = 5$ branches. The number of independent mesh equations is:

- ☐ 2 ☒ 3 ☐ 5 ☐ 8

Why: $m = B - N + 1 = 5 - 3 + 1 = 3$ (and there are $N - 1 = 2$ independent node equations: $2 + 3 = 5$ equations for the 5 unknown branch currents).

Three identical 30Ω resistors in parallel are equivalent to:

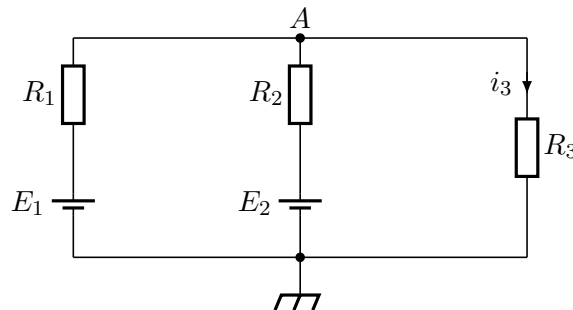
- ☐ 90 Ω ☐ 30 Ω ☒ 10 Ω ☐ 3.3 Ω

Why: $1/R_{\text{eq}} = 3/30 \text{ S}$, so $R_{\text{eq}} = 10 \Omega$.

Exercise 4. Millman's theorem

(5 marks)

In the circuit below, the ground (bottom wire) is the reference of potentials. Using Millman's theorem, determine the potential V_A of node A. Deduce the current i_3 that flows through R_3 towards the ground.



$$E_1 = 12 \text{ V} \quad R_1 = 2.0 \Omega \quad E_2 = 6.0 \text{ V} \quad R_2 = 3.0 \Omega \quad R_3 = 6.0 \Omega$$

The potentials at the far ends of the three branches are E_1 , E_2 and 0:

$$V_A = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{0}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$\text{Numerator: } \frac{12}{2.0} + \frac{6.0}{3.0} + 0 = 6.0 + 2.0 = 8.0 \text{ A} \quad \text{Denominator: } \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = 1.0 \text{ S}$$

$$V_A = 8.0 \text{ V} \quad i_3 = \frac{V_A - 0}{R_3} = \frac{8.0}{6.0} \text{ A} \quad i_3 = 1.3 \text{ A} \quad (4/3 \text{ A})$$

Check (node law at A): $i_1 = \frac{E_1 - V_A}{R_1} = 2.0 \text{ A}$ enters A, $\frac{E_2 - V_A}{R_2} = -0.67 \text{ A}$, and $2.0 - 0.67 = 1.33 \text{ A} = i_3$. ✓

Marking

①		
0	1	
②		
0	1	
③		
0	1	
④		
0	1	
⑤		
0	1	

Marking scheme

- ① **1 pt** Millman's formula written literally, with the right potentials (E_1 , E_2 , 0).
- ② **1 pt** Numerator = 8.0 A.
- ③ **1 pt** Denominator = 1.0 S.
- ④ **1 pt** $V_A = 8.0$ V.
- ⑤ **1 pt** $i_3 = V_A/R_3 = 1.3$ A.

Examiner's tip

Millman gives V_A as a mean of the potentials weighted by the conductances. The result must lie between the smallest and the largest potential (0 and 12 V): a quick sanity check.

Your score: / 14 $\longrightarrow$ mark out of 20 = score $\times$ 20/14 = / 20